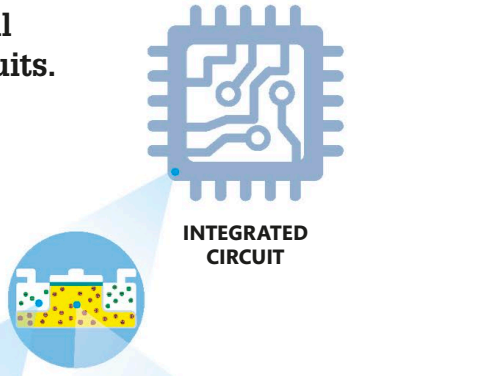
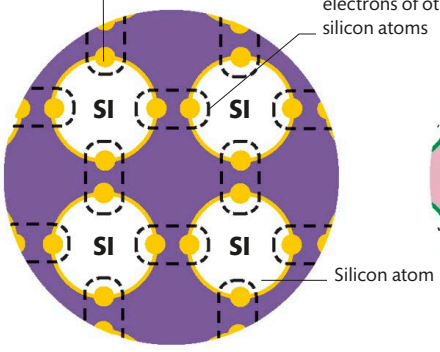
Digital electronics

Inside digital devices, information is processed by transistors—electronic components etched onto small pieces of semiconductor material—in integrated circuits.

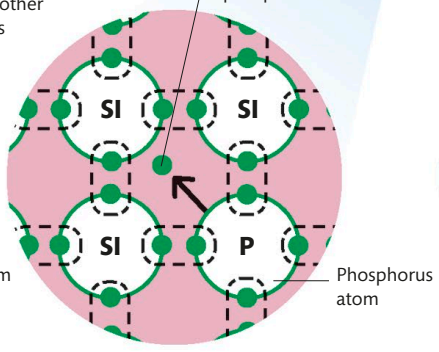


Semiconductors

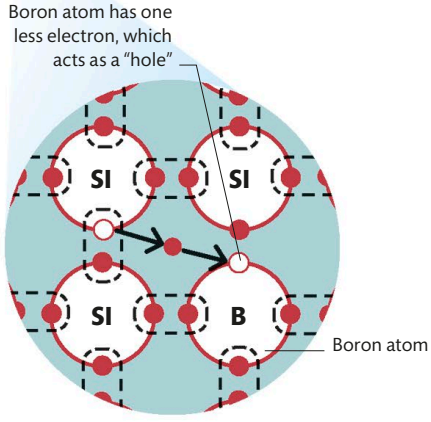
Materials called semiconductors are at the heart of the digital world. The most common is the element silicon. Pure silicon is not a very good conductor of electricity, but adding impurities of other elements, or "doping," enables it to conduct an electric current, which is a flow of electric charge. By adding different elements to a semiconductor, its distribution of positive and negative charges can be precisely controlled to direct a current through it.



Silicon
Pure silicon can conduct electricity only when heat or light gives electrons enough energy to break free of their atoms.



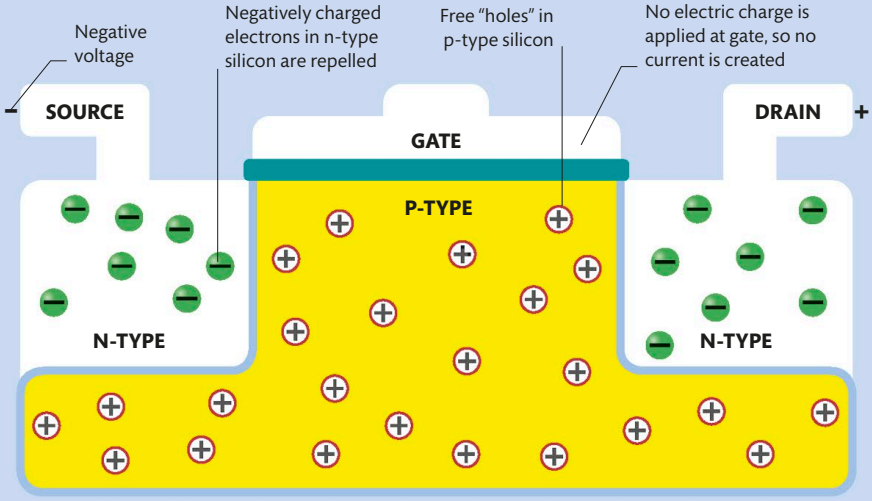
N-type (negative) silicon
Adding phosphorus atoms makes an n-type semiconductor with negatively charged electrons that are free to move.



P-type (positive) silicon
Adding boron means there are not enough electrons. This leaves positively charged "holes" that can move through the silicon.

Transistors

The transistors found on integrated circuits are made from pure silicon doped precisely to produce n- and p-type regions. Current can flow through a transistor, from "source" to "drain," only if there is an electric field applied to a part called the "gate." Current flowing represents a binary "1"; no current, a "0."



Transistor is "off"
The source is connected to a negative voltage, pushing electrons toward the drain. But only "holes"—not electrons—can flow through an adjacent area of p-type silicon.